CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 1: Multithreading & Concurrency Synchronization

Course: Advanced Java Programming [23CS3501]
Category: Assignments | Assignment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Implement robust thread synchronization, lock mechanisms, and deadlock prevention.

2. Assignment Problems & Instructions:

1. Explain thread lifecycle states and how Thread.sleep differs from Object.wait .
2. Implement a synchronized bounded-buffer Producer-Consumer queue in Java.
3. Demonstrate ReentrantLock and Condition variables compared to synchronized blocks.
4. Write a program causing and detecting thread deadlock using ThreadMXBean.

Submission Guidelines & Verification:

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